IMPERIAL COLLEGE LONDON

EE4-14
EE9-SO16

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

SPEECH PROCESSING

Thursday, 14 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.A. Naylor
Second Marker(s) : W. Dai

© Imperial College London

Speech Processing

1. Let $s(n)$ be a speech signal with discrete time index n and with sample values distributed according to a continuous probability density function $p(s)$. Now consider a process $q(s)$ that quantizes $s(n)$ to give a quantized version, $s_q(n)$, using uniform quantization with N quantization levels s_i for $i = 1, 2, ..., N$.

   a) i) Explain whether uniform or non-uniform quantization is more advantageous for speech signals, and give reasons.

   ii) State the main differences between uniform quantization and mu-law. [2]

   b) Sketch labelled illustrative histograms of the samples of quantization noise obtained when uniform quantization is applied to a typical speech signal using

   i) $N = 8$,

   ii) $N = 256$. [4]

   c) Derive an expression for the mean square quantization noise in $s_q(n)$. [4]

   d) Show that the mean square quantization noise is minimized when

$$\int_{(s_{i-1}+s_i)/2}^{(s_i+s_{i+1})/2} (s - s_i) p(s)\, ds = 0.$$

   [4]

   e) Consider the case of $N = 3$ with

$$p(s) = \begin{cases} 1 - |s| & \text{for } |s| \leq 1, \\ 0 & \text{for } |s| > 1. \end{cases}$$

   Find the quantization levels s_i giving minimum mean square quantization noise. [6]

 © 2020 Imperial College London

2. a) Let r_k for $k = 1$ to N denote the reflection coefficients in the lossless tube model of the speech production system, and r_G denote the reflection coefficient at the glottis.

i) Show that the transfer function of this model is given by

$$V(z) = \frac{0.5(1+r_G)\prod_{k=1}^{N}(1+r_k)z^{-N/2}}{D(z)}.$$

Give full details of your derivation. [2]

ii) Also derive an expression for the denominator $D(z)$ in terms of the reflection coefficients. [1]

b) Now consider the case when $r_G = 1$. In this case, $D(z)$ can be written $D(z) = \mathbf{P}_N \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, where bold typeface indicates vectors and matrices.

i) What would $r_G = 0$ imply for the physical system represented by the lossless tube model? Include an explanation in terms of characteristic acoustic impedance. [3]

ii) Find a recursive expression for $\mathbf{P}_k$ in terms of $\mathbf{P}_{k-1}$ for $k = 1, \ldots, N$. [2]

iii) Hence deduce a recursive expression for $D_k(z)$ in terms of $D_{k-1}(z)$. [2]

iv) Exploit the recursive expression to write out $D_1(z)$, $D_2(z)$ and $D_3(z)$. [2]

c) Ignoring latency, draw a labelled signal flow graph corresponding to the lossless tube model of order 3 using delays corresponding only to half a sampling period. [3]

d) Using the recursion of part (b) in reverse order, find an expression for $D_{k-1}(z)$ in terms of $D_k(z)$, clearly indicating the range of k. [5]

3. a) A speech signal s is processed to give

$$S(k,m) = \sum_{i=0}^{N-1} w(i)s(m-i)\exp\left(-j\frac{2\pi}{N}k(m-i)\right).$$

i) Explain this processing in overview.

ii) Define m, k and w.

iii) Explain the purpose of the term $(m-i)$ in the exponent.

iv) Explain what is meant by frequency resolution in this context, and indicate how the frequency resolution can be determined.

v) For any particular m, how many terms of $S(k,m)$ are independent?

vi) Consider a particular value of k. Explain the effect of w. Support you explanation with any appropriate formulae and/or diagrams. [7]

b) In a real-world application of speech processing, a signal model is used comprising $s(n)$ representing a speech signal and $\nu(n)$ representing additive noise. The signal model is written in the time domain as $y(n) = s(n) + \nu(n)$ and in the frequency domain as $Y(l,k) = S(l,k) + V(l,k)$ for time-frame index l and frequency index k.

i) Explain what is meant by spectral variance of a signal in this context. [2]

ii) Write down the mathematical definition for the spectral variance of $y(n)$, denoted $\phi_y(l,k)$. [2]

c) Consider a device which performs noise reduction on a noisy speech signal. The device uses the method of amplitude spectral subtraction.

i) Show that the output signal of the device at time-frame l and frequency index k can be written

$$Z(l,k) = H(l,k)Y(l,k)$$

and write down an expression for $H(l,k)$ in terms of $\hat{\phi}_v(l,k)$. [4]

ii) Describe and explain the minimum statistics method for determining the estimate $\hat{\phi}_v(l,k)$. State the assumptions that are required for this estimation method to be accurate. State any specific advantages of the minimum statistics method. [5]

4. a) i) Using a few sentences and illustrative examples, explain the difference between a phone and a phoneme. [2]

ii) Explain how an automatic speech recognizer could be designed to be robust to coarticulation. [2]

b) An example of an isolated word speech recognition system uses S states for each word. A hidden Markov model (HMM) is applied for which the transition probability from any state i to next state j is denoted a_{ij}. The speech signal to be recognized is segmented into frames with each frame represented by a feature vector $\mathbf{x}_i$ for $i = 1, 2, \ldots, T$.

i) Describe an appropriate training procedure for this HMM.

Include in your answer a definition of what data is required by the training procedure, any relevant mathematical analysis and supporting diagrams. [5]

ii) For a particular state q in the HMM, let the probability of a transition from state q to the next state be denoted p. Derive an expression for the average time duration (measured in frames) spent in state q. Show your working. [4]

iii) A simplified speech recognition system is designed using an HMM with 3 states. The state diagram of the HMM is shown in Figure 4.1 in which the labels on the arrows indicate the state transition probabilities.

In an experiment, this speech recognition system receives as input the feature vectors

$$\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_5.$$

Table 1 shows the output probability densities of the feature vectors for each state.

Let

$$B(s,t)$$

be the maximum probability density that the model generates the sequence of feature vectors $\{\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_t\}$ from any sequence of states for which frame 1 is in state 1 and frame t is in state s.

Construct a lattice diagram showing all the feasible paths from frame 1 being in state 1, denoted $(1,1)$, to frame 5 being in state 3, denoted $(3,5)$.

Determine the maximum probability alignment of the feature vectors to the states of the HMM and the value of $B(3,5)$ for this alignment. [7]

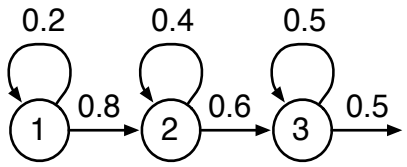


Figure 4.1: Hidden Markov model.

	frame $\mathbf{x}_1$	frame $\mathbf{x}_2$	frame $\mathbf{x}_3$	frame $\mathbf{x}_4$	frame $\mathbf{x}_5$
state 1	0.5	0.2	0.6	0.4	0.5
state 2	0.5	0.7	0.3	0.1	0.5
state 3	0.5	0.5	0.8	0.6	0.6

Table 1: Output probabilities.